

# iTrace aberrometry - Identifying occult imperfections in the visual system

## ABSTRACT

Visual acuity is the sum of qualitative and quantitative factors. A combination of both determines the final visual acuity. The quality of vision (QoV) can be easily assessed by documenting objectively the higher-order aberrations or by using subjective questionnaires available. The QoV is an integration of varied optical and neural factors. Similarly, the quantity of vision is documented by measuring uncorrected and best-corrected distance visual acuities. Optical aberrations are defect in a lens or a mirror prevents light rays from being focused at a single point and results in a distorted or blurred image. Moreover, they are the departure of the performance of an optical system from the predictions of paraxial optics. The QoV is primarily affected by both higher- and lower-order optical aberration. Aberrometers are the most vital instruments used for estimating optical aberrations so that a more comprehensive understanding of optical error can be quantified and corrected. A variety of aberrometers with different principles are available, such as Ray Tracing, Hartmann-Shack, Tscherning, and automatic retinoscopy. In this review, our prime focuses on Ray Tracing aberrometer, iTrace. This review will help all the ophthalmologists including residents and fellows learn the principle, features, and clinical applications of iTrace. The system integrates corneal topography with wavefront aberrometry, which has the unique feature of revealing the internal aberrations of the eye by subtracting the corneal aberrations from total aberration. It is the investigation of choice in the today's era considering patient satisfaction and visual outcomes, postpremium intraocular lens implantations.

**Keywords:** Corneal topography, iTrace, Ray Tracing aberrometer, wavefront aberrometry

## INTRODUCTION

Human visual acuity is a sum of visual quality and quantity. Any deviation from these is qualified as poor visual acuity.<sup>[1]</sup> This can result from the poor quantity of vision which is calculated as uncorrected and best-corrected visual acuity. The quality of vision (QoV) is documented objectively by calculating the higher-order aberrations (HOAs) or by using subjective questionnaires available.<sup>[2]</sup> Visual acuity is primarily determined by tear film dynamics, cornea aberrations, pupil size, lenticular aberration, and retinal pathology.<sup>[3]</sup> The first three are the major causes of optical aberrations. Optical aberrations are one of the major causes of poor QoV. Optical aberrations are synonymous with refractive errors. Most of us are familiar with myopia, hyperopia, and astigmatism, which are labeled as a lower-order aberration (LOA).<sup>[4]</sup> In routine practice, optometrists have been correcting LOAs for over a

century with spectacles. However, what about HOAs such as coma, trefoil, and secondary astigmatism. They cannot be corrected by spectacles and are a major source of patient dissatisfaction considering the QoV.<sup>[5]</sup> The aberrometers are intended to measure the deviation of rays at the retinal plane of focus when the light rays pass through the pupil.<sup>[6]</sup> There are a variety of aberrometers available in the market in the form of Hartmann-Shack, Tscherning, Ray Tracing, and automated aberrometers. They help in diagnosis and

## BHARAT GURNANI, KIRANDEEP KAUR<sup>1</sup>

Department of Cataract, Cornea and Refractive Services,

<sup>1</sup>Department of Cataract, Pediatric Ophthalmology and Strabismus Services, Aravind Eye Hospital and Post Graduate Institute of Ophthalmology, Puducherry, India

**Address for correspondence:** Dr. Bharat Gurnani, Aravind Eye Hospital and Post Graduate Institute of Ophthalmology, Puducherry - 605 007, India. E-mail: drgurnaniibharat25@gmail.com

Submitted: 13-May-2021 Revised: 15-May-2021  
Accepted: 15-May-2021 Published: 08-Dec-2021

## Access this article online

### Website:

www.kjophthal.com

### DOI:

10.4103/kjo.kjo\_112\_21

### Quick Response Code



This is an open access journal, and articles are distributed under the terms of the Creative Commons Attribution-NonCommercial-ShareAlike 4.0 License, which allows others to remix, tweak, and build upon the work non-commercially, as long as appropriate credit is given and the new creations are licensed under the identical terms.

**For reprints contact:** WKHLRPMedknow\_reprints@wolterskluwer.com

**How to cite this article:** Gurnani B, Kaur K. iTrace aberrometry - Identifying occult imperfections in the visual system. Kerala J Ophthalmol 2021;33:373-83.

treatment of various optical pathologies in the eye.<sup>[7]</sup> iTrace is a Ray Tracing aberrometer that employs both wavefront (WF) aberrometry and Placido corneal topography (CT). It can calculate corneal, lens, and total aberration separately along with topographic screening.<sup>[8]</sup> Recently, it has gained immense popularity for preoperative and postoperative evaluation and management of cataract and refractive surgery patients.<sup>[8]</sup>

## BEFORE UNDERSTANDING ITRACE AS A WHOLE, WE MUST UNDERSTAND FEW IMPORTANT TERMINOLOGIES

### Wavefront

It is a line drawn perpendicular to all rays of light entering/exiting the eye. The WF is flat or plane WF until it hits the curved surface of the eye. The more distorted the optical system, the more distorted the WF. The parallel beams have a parallel WF, while the converging beams result in a spherical WF.<sup>[9]</sup>

### Aberrations

They are imperfections in the image formation of the optical system. In other words, they are also defined as a defect in the lens or the focusing mirror that prevents light rays from being focused at a single point and results in a distorted or blurred image. They are also defined as the difference in optical path length between any ray passing through the pupillary plane and the chief ray passing through the pupillary center.<sup>[10]</sup>

## TYPES OF ABERRATIONS

### Diffraction

Bending of light rays caused by the edge of an aperture of the rim of a lens. The actual pattern of the diffracted image produces a circular aperture or pupil is a series of concentric bright and dark rings.

### Chromatic aberration

Chromatic aberration, also called “color fringing” or “purple fringing,” is a common optical aberration that occurs when a lens is either unable to focus all wavelengths of color to the same focal plane and/or when wavelengths of different colors are focused at different positions in the focal plane. It is a result of lens dispersion, with different colors of light traveling at different speeds while passing through a lens. As a result, the image usually looks blurred, or noticeable colored edges (red, green, blue, yellow, purple, and magenta) can appear around objects, especially in high-contrast situations.

### Spherical aberration

It is an optical problem that occurs when all parallel rays of light focus at different points after passing through a spherical surface. The ray of light passing through closer to

the edge or “periphery” of the lens is refracted more than near its horizontal axis. Hence, they end up in different spots across the optical axis. These aberrations affect the resolution and clarity, making it difficult to obtain sharp images.

### Geometric aberration

They are caused by the change in the geometry, i.e., shape of the lens and mirror. They are also called monochromatic aberrations because they occur for images formed with the light of a single frequency. Common examples include coma, distortion, decentering, astigmatism, and field curvature.

### Oblique aberration

Objects in the peripheral field are seen by the oblique incident narrow pencil of rays, which are limited by the pupil. Thus, refracted pencil show oblique astigmatism.<sup>[11]</sup>

## CLASSIFICATION

1. LOAs - Myopia, hyperopia, and regular astigmatism
2. HOAs - Spherical aberrations, coma, decentering, chromatic aberrations, oblique astigmatism, and centering.<sup>[4]</sup>

## WAVEFRONT ABERROMETRY

WF aberrometry is an objective method of measuring the refractive power of the eye. The aberrometers give detailed measurements and interpretation of the movement of the light WF. When parallel rays of light pass through the pupil and are reflected by the retina, the WF of the reflected beam is recorded and compared to that of the reference WF. All the points of variation are recorded and measured, creating in turn an optical fingerprint. It assists in the diagnosis of both higher-order and lower-order refractive errors.<sup>[12]</sup>

## DIFFERENT TYPES OF ABERROMETER

Types of WF testing:<sup>[13]</sup>

1. Ingoing WF aberrometry - Retinal image aberrometry
2. Outgoing WF aberrometry - Hartmann-Shack sensor - e.g., Zywave
3. Double-pass aberrometry - Automated slit skiascopy
4. Ingoing subjective adjustable aberrometry - Spatially resolved refractometry
5. Tscherning's aberrometer - Tracey retinal ray tracing - e.g., NIDEK OPD Scan

## RAY TRACING – ITRACE

### Concept of Ray Tracing

In this method, the parallel laser beam passes across the line

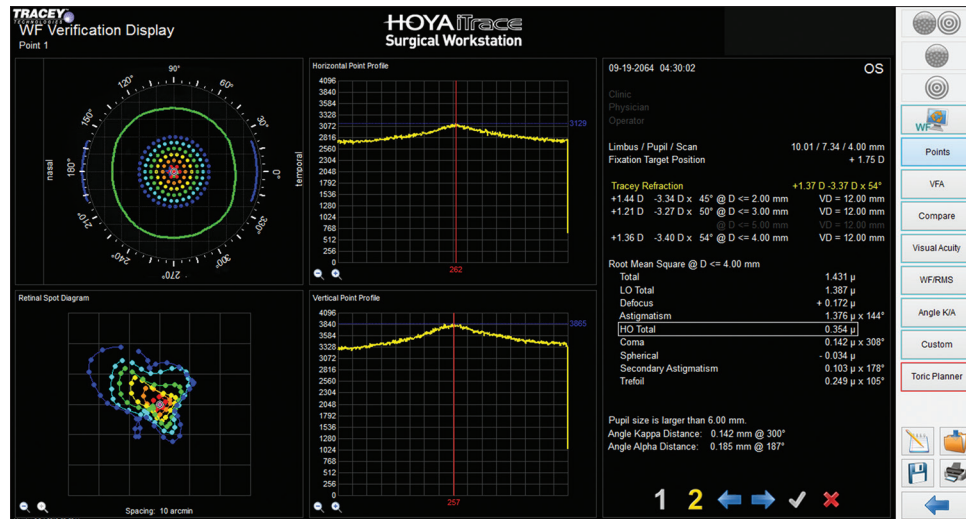


Figure 1: Image of the iTrace depicting the wavefront verification display along with retinal spot diagram pattern. The vertical point profile is slightly higher than the horizontal point profile, but the pattern is uniform

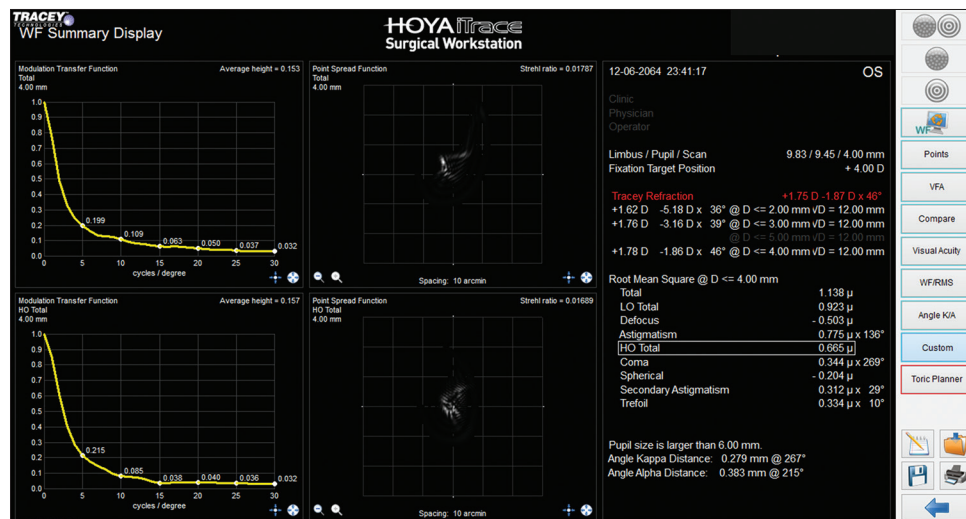


Figure 2: Image of the iTrace depicting defocus suggestive of refractive error

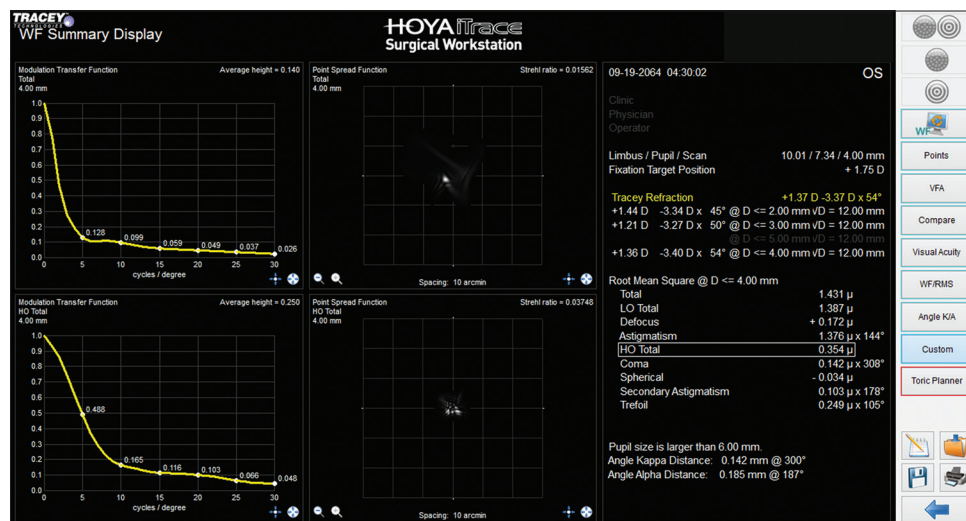


Figure 3: Image of the iTrace depicting modulation transfer function of a patient with refractive error. Note the slope will be uniform in an emmetropic patient

of sight through the pupil. The aberrometer employs the retroreflected light captured by reference linear sensors to measure the exact location where the laser beam strikes the retina. The local aberrations that are present in the path of the laser beam and that pass through the cornea and internal structures cause a shift in the location of the retina. Thus, laser beam is shifted to another point, and thus, multiple points are projected onto the retina. These laser beams pass through multiple points on the pupil. By this process of entrance and exit of laser beams, a real WF error is obtained. This method helps obtain the forward aberrations of light that go through the pupil.<sup>[14]</sup>

### UNDERSTANDING RETINAL SPOT DIAGRAM, POINT SPREAD FUNCTION, AND MODULATION TRANSFER FUNCTION

#### Retinal spot diagram

When several laser points pass through the pupil and are projected on the retina, a retinal spot diagram (RSD) is created. The RSD reveals the patient's refractive error, aberration, and data of point spread function (PSF) [Figure 1].<sup>[15]</sup>

#### Point spread function

It simply means the spread of points on the retina once the spots are projected through the pupil. The morphological pattern obtained gives an idea of the degree of WF qualitative aberrations. The smaller and sharper the PSD, the better the resolution [Figure 2].<sup>[15]</sup>

#### Modulation transfer function

It describes how details of the objects are obtained in the form of the image produced by the lens. In other words, the modulation transfer function (MTF) and PSF define the optical system's ability [Figure 3].<sup>[15]</sup>

### PRINCIPLE OF ITRACE

iTrace employs the basic principle of ray tracing as described above. Here, series of infrared beams (100  $\mu\text{m}$ ) of 785 nm wavelength pass parallel through the pupil in the line of sight and project on the retina. Once this position has been obtained, the laser beam is shifted sequentially to another point on the retina. In this way, 64 laser beams are projected through the entrance pupil 4 times each (256 points) at a rapid speed of 250 ms. In this way, a pattern is obtained RSD (map) of all the projected points, and a WF is reconstructed revealing the occult aberrations. In an emmetropic system, normally, all the points will concentrate over a single focus macula. Any local aberrations when the beam enters the cornea or the lens cause a shift in the location of the retina, concerning the reference focus points. The major advantage of iTrace over other aberrometers is that other rectilinear or polar patterns can also be obtained. The iTrace uses concentric rings to depict the pattern.<sup>[15]</sup>

### COMPONENTS

The panel and the components of iTrace are collectively labeled as the data acquisition unit. Through this, the laser beams are projected onto the retina and then processed through the inbuilt software. The unit has an inbuilt optometer, pupil size detector, and adjustable focus. The optometer helps by aligning the patient pupillary axis to the laser beam projection axis and simultaneously helps by relaxing the accommodation of the patient by using the principle of fogging (+7D to -5D). In addition, it has a corneal topographer to map the cornea [Figure 4].

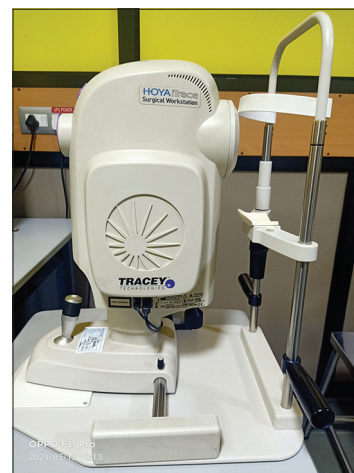
### TECHNICAL PREREQUISITES TO BE FULFILLED TO GET A CORRECT GRAPH

Specifications of the i-Trace system: <sup>[16]</sup>

1. Refractive power range:  $\pm 15$  D sphere,  $\pm 10$  D cylinder
2. Pupil diameter: 2.5–8 mm
3. Accuracy:  $\pm 0.10$  D
4. Reproducibility:  $\pm 0.10$  D
5. Topographic cone 24 rings
6. Measured and analyzed points: 8640
7. Corneal coverage: 8 mm on 42.5D sphere
8. Dioptric range: 33.75–61.36D
9. Capture mode: auto and manual
10. Dimensions: 13 inch (33 cm)  $\times$  17 inch (43.2 cm)
11. Weight: 27.4 lb (20 kg).

### INTERPRETATION OF RESULTS

iTrace provides measurements in the form of few basic maps. These maps are described below<sup>[17]</sup>



**Figure 4:** Image of the iTrace depicting the data acquisition unit of iTrace measurement system which combined the features of wavefront aberration along with Placido-based topography

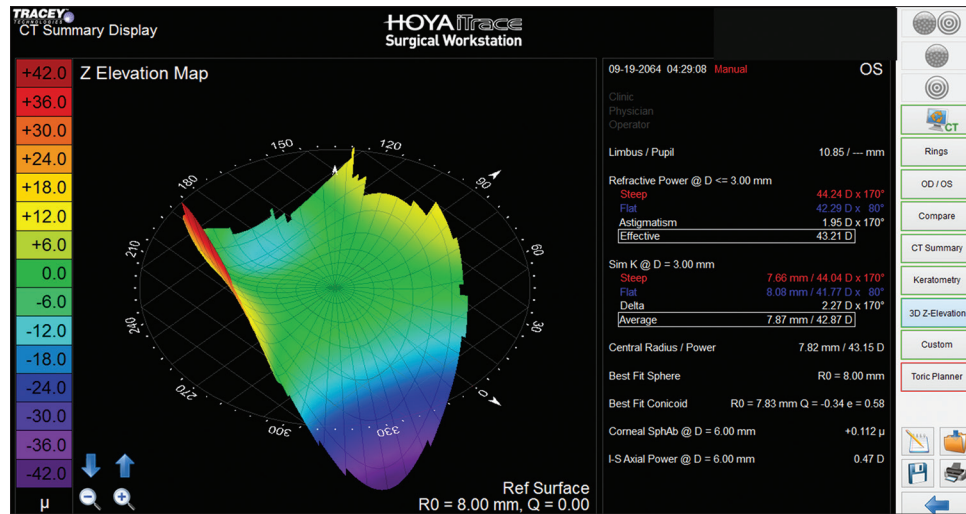


Figure 5: Image of the iTrace depicting the corneal topography summary display showing Z elevation map

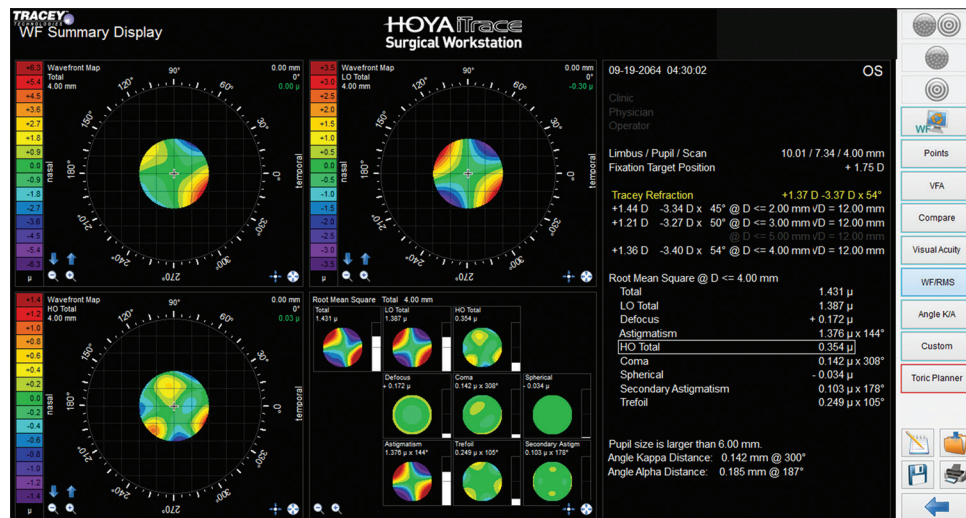


Figure 6: Image of the iTrace depicting the wavefront verification display with root mean square map of all the aberration of the patient

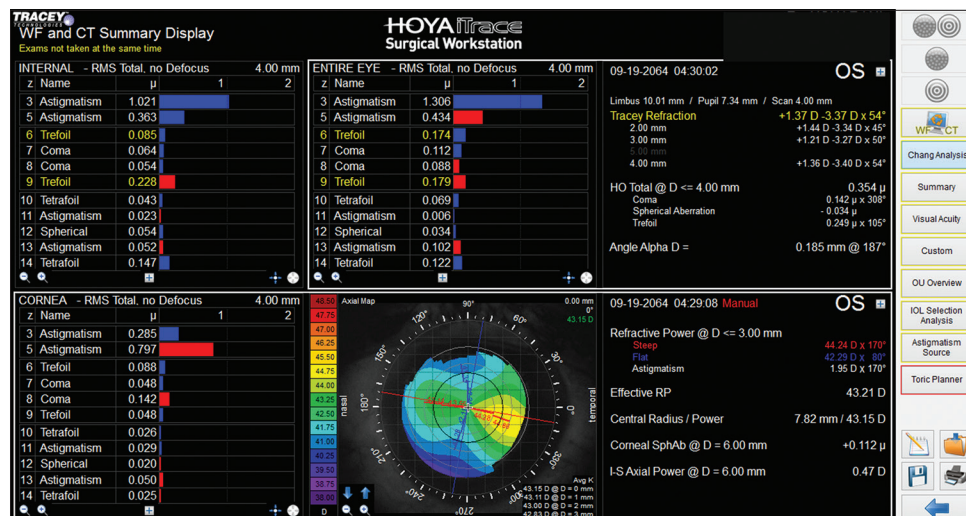


Figure 7: Image of the iTrace depicting the total and higher-order aberrations of the patient



Figure 8: Image of the iTrace depicting blurred image of Snellen E with an acuity of 20/70. The blurred vision is a combination of corneal and lenticular aberrations



Figure 9: (a) Image of the iTrace depicting corneal topography summary display showing the wavefront map, Z elevation map, local ROC map, and refractive map. (b) Image of the iTrace depicting the keratometry map

### Wavefront map (total and higher-order aberration)

This map reveals the WF aberration which is color coded and values are measured in microns. The total map gives both HOA and LOA, and HOA reveals higher aberrations such as coma, trefoil, and spherical aberration. Warm colors indicate that the WF aberrations are in front of the reference plane while cooler colors indicate that the WF aberrations are at the back of the reference plane. For example, red indicates warmer, green indicates neutral, and blue indicates cooler colors [Figure 5].

### Root mean square

The root mean square (RMS) is the quantitative analysis of the magnitude of the aberration. This gives a total RMS value for the total ocular aberration of the eye and a specific RMS

value for each Zernike term or component of the eye. The measurements are taken in microns and up to sixth-order aberration can be assessed [Figure 6].

### Total refractive and higher-order aberration refractive map

In this map, the refractive power of the eye is depicted in diopters. Myopia is denoted by red, emmetropia in green, and hypermetropia in blue. The measurement taken is for the whole eye rather than corneal power alone. The combination of the refractive and the topographic map display gives an idea of whether the astigmatism is corneal or lenticular. In addition, accommodation can also be measured by obtaining the maps at various distances and assessing the refractive change [Figure 7].

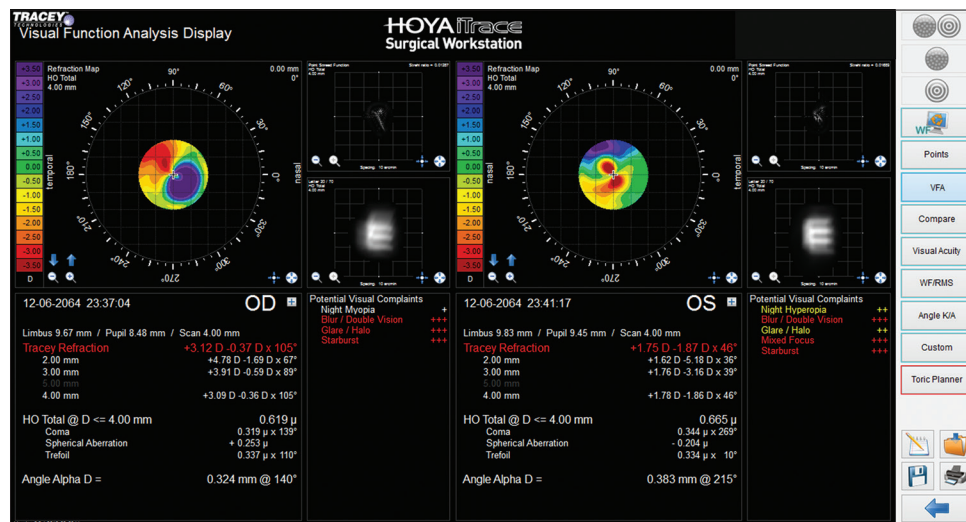


Figure 10: Image of the iTrace depicting the visual function analysis display comparison of the two eyes showing the potential visual complaints such as glare, halo, and starburst

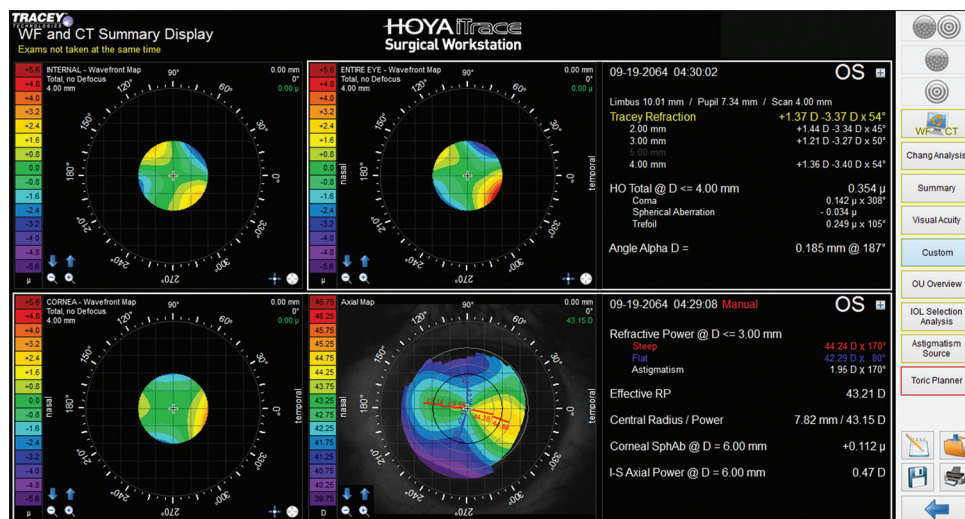


Figure 11: Image of the iTrace depicting combined wavefront and corneal topography summary display

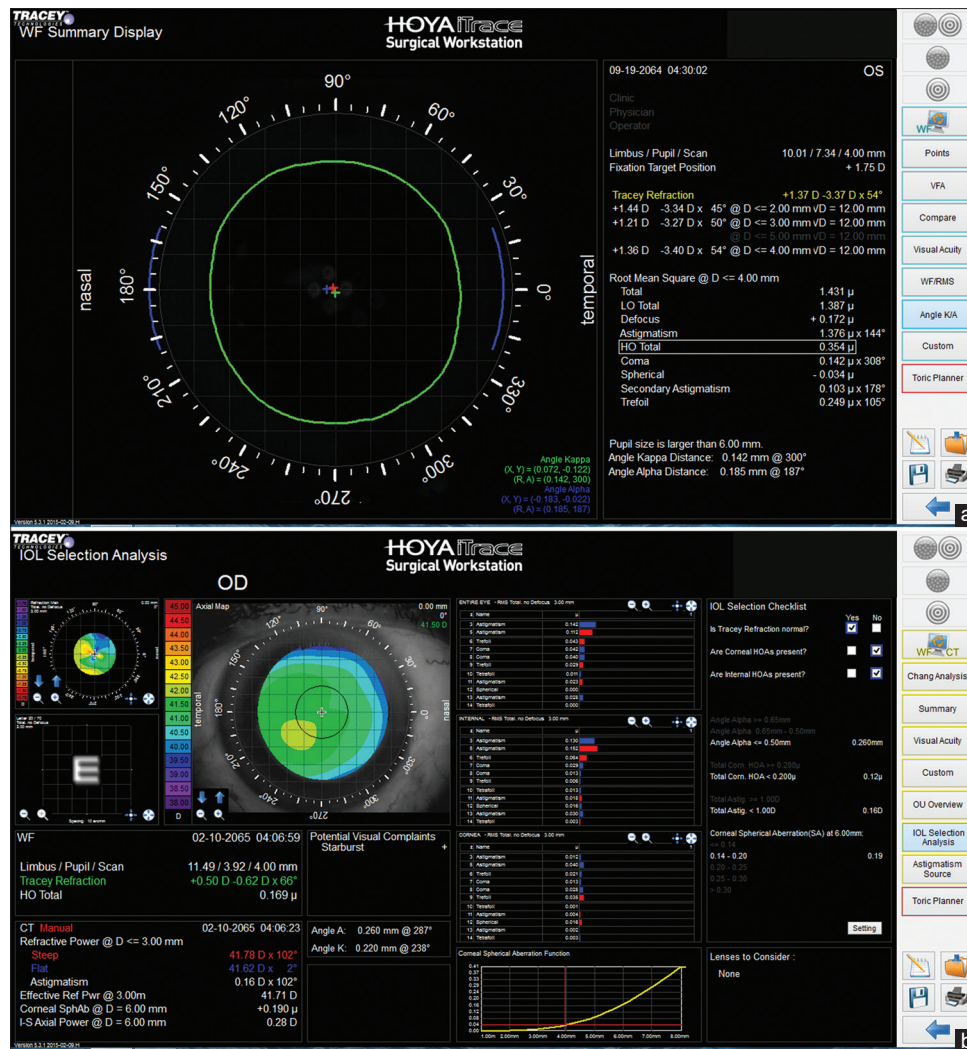


Figure 12: (a) Image of the iTrace depicting the angle kappa for premium intraocular lens analysis. (b) Image of the iTrace depicting the intraocular lens selection analysis based on various aberrations

### Point spread function - Total and higher-order aberration

It represents the quality and sharpness of the image obtained in the optical system. When point source of light passes through the pupil and strikes the retina, the morphological pattern is obtained is the PSF. The higher the aberrations and refractive error, the greater is the defocus. An emmetropic eye gives a well-centered focus [Figure 2].

### Snellen letter - Total and higher-order aberration

This map is a measure of how the human eye will see the letter “E” at various grades of visual acuity 6/6, 6/12, 6/24, 6/36, and 6/60. This helps the ophthalmologists or optometrists to assess how are the patient’s QoV and any discomfort reported by the patient [Figure 8].

### Modulation transfer function

This helps obtain images produced by the optical system in terms of details of the object. This function is a measure

of contrast sensitivity and resolution of an image at various spatial frequencies [Figure 3].

### Zernike polynomials

This map displays the Zernike polynomials (ZPs) giving a detailed analysis of specific aberrations in the eye. The iTrace only displays ZP up to sixth order with 27 terms, with total internal aberrations, corneal, as well as the difference between the cornea and total of the eye.

### Aberration of internal optics

This map and property are unique to the iTrace system which provides important information combining WF analysis (WFA) with CT. CT maps mathematically generate the corneal aberration. These aberrations when subtracted from the total aberrations give the internal aberrations of the optical system. This features to distinguish and separate the corneal and internal optical aberrations.

## CORNEAL TOPOGRAPHY

The iTrace has an inbuilt Placido (Vista) topographic system (Eyesys Vision Inc., Houston, TX, USA) to assess the corneal aberrations. The system covers the central 0.6 mm and up to 10 mm of the peripheral cornea.

The topography system helps to analyze:

1. K reading (3 mm zone)
2. Refractive corneal power (3 mm zone)
3. Corneal indices - Inferior-superior asymmetry, potential corneal acuity, aspheric factor-quality index (Q), uniformity index, and spherical aberrations of cornea
4. Topographic map display - Axial map, tangential map, elevation map, refractive map, and WF map [Figure 9a and b].

## DATA ANALYSIS

1. WFA - There are two displays.
  - a. Visual function analysis (summary display) shows the total HOA either the refraction map or the WF map. This display gives a detailed insight into refraction (various diameters), RMS, PSF, Snellen letters, and symptoms such as myopia, glare, and halos. The symptoms are graded 1–3 depending on the intensity, and the pupil diameter can also be altered with the help of the regulator at the bottom [Figure 10]
  - b. WF comparison map gives the display of two maps in a single patient. First, the refractive map for comparing aberration preoperatively and postoperatively, and second, the accommodation map.
2. Corneal topographic analysis (CT summary display) has been discussed in the respective section

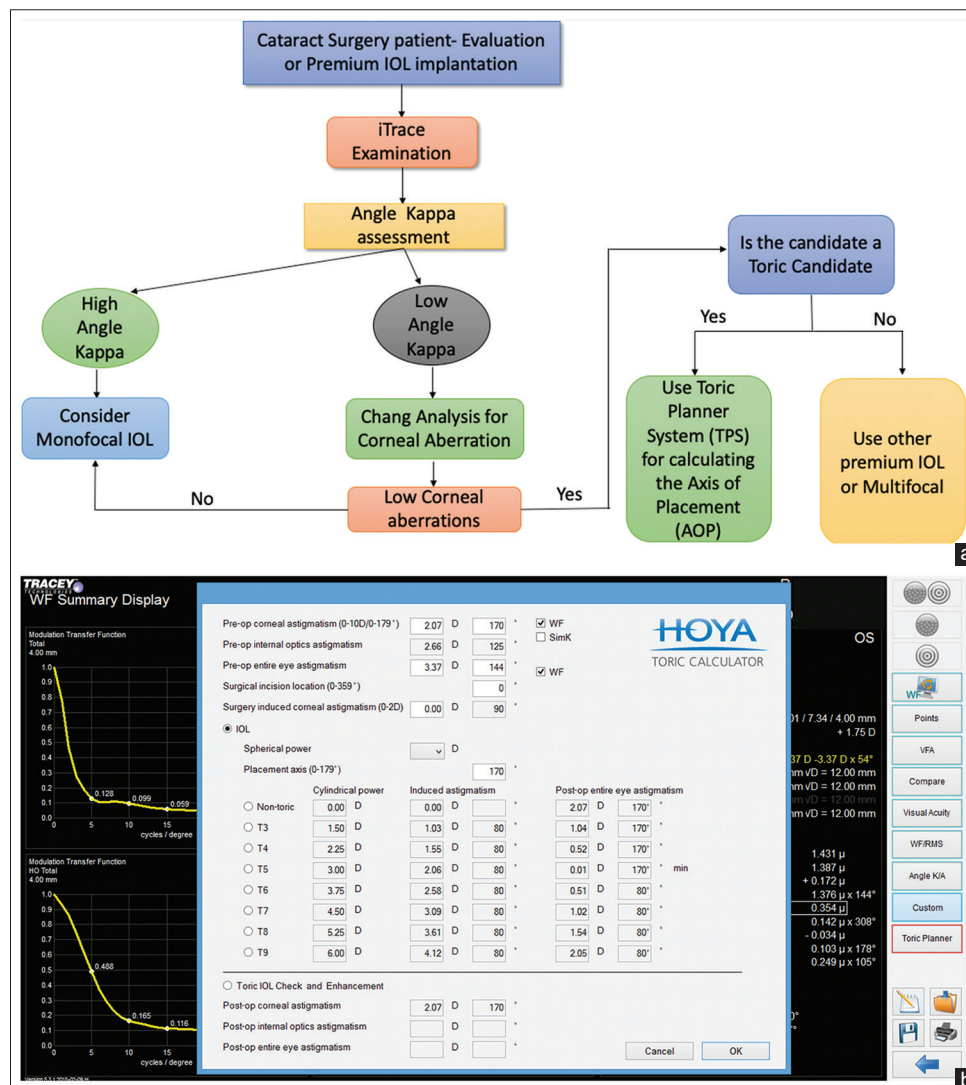


Figure 13: (a) Flowchart for premium intraocular lens selection. (b) Image of iTrace depicting the inbuilt toric planning system having the toric calculator for intraocular lens power and axis placement

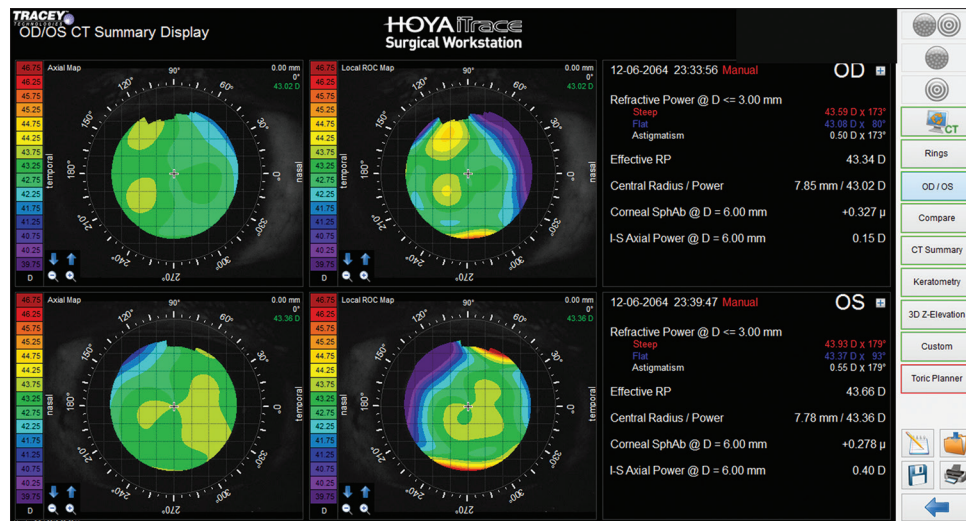


Figure 14: Image of iTrace depicting the comparison of both eyes corneal topography summary display

3. Combined WF and topographic (CT) analysis (summary display) [Figure 11].

#### CLINICAL APPLICATION

1. Aberration - WF verification with the help of RSD and RMS WF values allows rapid diagnosis of LOAs and HOAs
2. Visual function, quality, and symptoms - The iTrace gives the summary of multizone refraction analysis, PSF analysis, Snellen E simulation analysis, and potential visual complaints
3. The iTrace has the unique feature to separate corneal and lenticular aberrations
4. CT monitors the corneal changes between examinations and identifies the absolute effect of contact lenses, refractive surgery, and corneal warpage
5. Accommodation - Identify changes in accommodation up to 4D
6. Visual acuity analysis - Preoperative and postoperative total visual acuity changes after cataract and refractive surgery, contact lenses fitting, or spectacles usage
7. Keratoconus screening - Using CT maps and by employing Rabinowitz criteria [Figure 9a and b]
8. Contrast sensitivity helps in the assessment of CS based on MTF, RMS, and PSF
9. Choosing premium intraocular lens (IOLs) such as multifocal and toric IOL based on an assessment of angle Kappa. A high angle Kappa is a contraindication for implantation of premium IOLs [Figures 12a, b and 13a, b]
10. It also helps in postoperative of IOL in the form of IOL tilt and centration
11. The early cataract changes are reflected by the lens aberrations

12. The internal aberrations can also result from the posterior capsular. Opacification can be reconfirmed by slit-lamp examination
13. It can also display characteristics of both eyes simultaneously for comparison [Figure 14]. The other uses include corneal inlay planning, dry eye management, vitreolysis, and dysfunctional lens syndrome.<sup>[15-17]</sup>

#### FUTURE DIRECTION

iTrace besides being a diagnostic and therapeutic tool has a big future, considering research globally. It will also serve as an educational tool for patients, ophthalmologists, and trainees with a variety of inbuilt technical features.

#### Acknowledgments

Aravind Eye Hospital and Post Graduate Institute of Ophthalmology, Pondicherry, is acknowledged.

#### Financial support and sponsorship

Nil.

#### Conflicts of interest

There are no conflicts of interest.

#### REFERENCES

1. McKean-Cowdin R, Varma R, Wu J. Los Angeles Latino Eye Study Group. The severity of visual field loss and health-related quality of life. *Am J Ophthalmol* 2007;143:1013-23.
2. Hiraoka T, Okamoto C, Ishii Y, Kakita T, Okamoto F, Oshika T. Time course of changes in ocular higher-order aberrations and contrast sensitivity after overnight orthokeratology. *Invest Ophthalmol Vis Sci* 2008;49:4314-20.
3. Denoyer A, Rabut G, Baudouin C. Tear film aberration dynamics and vision-related quality of life in patients with dry eye disease. *Ophthalmology* 2012;119:1811-8.

4. Kligman BE, Baartman BJ, Dupps WJ Jr. Errors in treatment of lower-order aberrations and induction of higher-order aberrations in laser refractive surgery. *Int Ophthalmol Clin* 2016;56:19-45.
5. Karimian F, Feizi S, Doozande A. Higher-order aberrations in myopic eyes. *J Ophthalmic Vis Res* 2010;5:3-9.
6. Thompson KP, Staver PR, Garcia JR, Burns SA, Webb RH, Stulting RD. Using InterWave aberrometry to measure and improve the quality of vision in LASIK surgery. *Ophthalmology* 2004;111:1368-79.
7. Visser N, Berendschot TT, Verbakel F, Tan AN, de Brabander J, Nuijts RM. Evaluation of the comparability and repeatability of four wavefront aberrometers. *Invest Ophthalmol Vis Sci* 2011;52:1302-11.
8. Chen Y, Xia X. Comparison of the Orbscan II topographer and the iTrace aberrometer for the measurements of keratometry and corneal diameter in myopic patients. *BMC Ophthalmol* 2016;16:33.
9. Bruce AS, Catania LJ. Clinical applications of wavefront refraction. *Optom Vis Sci* 2014;91:1278-86.
10. Son HS, Łabuz G, Khoramnia R, Yildirim TM, Auffarth GU. Laboratory analysis and ray visualization of diffractive optics with enhanced intermediate vision. *BMC Ophthalmol* 2021;21:197.
11. Oshika T, Klyce SD, Applegate RA, Howland HC, El Danasoury MA. Comparison of corneal wavefront aberrations after photorefractive keratectomy and laser *in situ* keratomileusis. *Am J Ophthalmol* 1999;127:1-7.
12. He L, Manche EE. Contralateral eye-to-eye comparison of wavefront-guided and wavefront-optimized photorefractive keratectomy: A randomized clinical trial. *JAMA Ophthalmol* 2015;133:51-9.
13. Cade F, Cruzat A, Paschalis EI, Espírito Santo L, Pineda R. Analysis of four aberrometers for evaluating lower and higher order aberrations. *PLoS One* 2013;8:e54990.
14. Chen Y, Xia X. Comparison of the Orbscan II topographer and the iTrace aberrometer for the measurements of keratometry and corneal diameter in myopic patients. *BMC Ophthalmol*. 2016 Mar 31;16:33. doi: 10.1186/s12886-016-0210-8. PMID: 27029933; PMCID: PMC4815140.
15. Molebny VV, Panagopoulou SI, Molebny SV, Wakil YS, Pallikaris IG. Principles of ray tracing aberrometry. *J Refract Surg* 2000;16:S572-5.
16. Rozema JJ, Van Dyck DE, Tassignon MJ. Clinical comparison of 6 aberrometers. Part 1: Technical specifications. *J Cataract Refract Surg* 2005;31:1114-27.
17. Molebny VV, Pallikaris IG, Naoumidis LP et al. Retinal ray tracing technique for eye refraction mapping. *Proc SPIE* 1997; 2971: 175-183..